

MediReco

Personalized Healthcare & Medicine Recommendation System

Report date	15 September 2026, 12:45 PM	Report type	ML-assisted check-up summary
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Important notice

Educational decision support only. This report is not a diagnosis, prescription, treatment plan, or substitute for professional medical care.

1. Patient check-up inputs

Item	Value	Notes
Patient name	Demo Patient	optional PDF field
Age	32	years
Gender	male	
Blood pressure code	1	dataset scale 0-10
Cholesterol code	1	dataset scale 0-10
Fever	yes	
Cough	yes	
Fatigue	yes	
Difficulty breathing	no	
Pregnancy flag	No	context filter
Known allergy / medicine	None provided	user-entered

2. Model prediction summary

Rank	Predicted class	Model probability
1	Other	73.1%
2	Influenza	12.0%
3	Diabetes	5.8%

Risk model

Predicted risk class: **Medium** | Model probability: **52.4%**

Risk class	Probability
High	9.3%
Low	38.3%
Medium	52.4%

3. Treatment & care reference

Evidence-linked reference options for the top predicted classes. These are not prescriptions or dosing instructions.

Influenza - 12.0% model probability

Medicine / treatment references

Reference	Details	Safety / review
Oseltamivir	Antiviral used for some influenza infections; treatment is most useful when started early and it requires professional assessment.	Clinical review recommended

Other guidance

Acetaminophen: Can help reduce fever and aches; check the label and avoid duplicate acetaminophen-containing products.

Ibuprofen: May help relieve fever and body aches for appropriate patients; ask a clinician/pharmacist if you have kidney, stomach, heart or other conditions.

Fluids and rest: Rest and hydration can support recovery; seek care for severe or worsening symptoms.

Diabetes - 5.8% model probability

Medicine / treatment references

Reference	Details	Safety / review
Metformin	Commonly used oral medicine for type 2 diabetes; suitability depends on the individual and kidney/liver status.	Clinical review recommended
Other glucose-lowering medicines	Several medicine classes may be used alone or in combination depending on glucose control, kidney/heart health, side effects and other factors.	Clinical review recommended

Other guidance

Blood glucose monitoring: Regular glucose monitoring and individualized nutrition/activity planning are part of diabetes management.

4. Model explainability

Approximate Random Forest feature importance aggregated back to the original input fields. This shows which inputs influenced the trained model most in aggregate; it is not causal evidence.

Feature	Relative importance
Age	31.1%
Difficulty breathing	10.4%
Cholesterol code	10.3%
Gender	10.3%
Cough	10.0%
Fatigue	9.7%
Fever	9.1%
Blood pressure code	9.1%

5. Evidence sources used by the recommendation layer

MedlinePlus - <https://medlineplus.gov/druginfo/meds/a699040.html>

MedlinePlus - <https://medlineplus.gov/coldandcoughmedicines.html>

MedlinePlus - <https://www.medlineplus.gov/ency/article/000709.htm>

MedlinePlus - <https://medlineplus.gov/flu.html>

MedlinePlus - <https://medlineplus.gov/druginfo/meds/a696005.html>

MedlinePlus - <https://medlineplus.gov/ency/patientinstructions/000989.htm>

6. Model and project notes

Algorithm: Random Forest Classifier

Number of trees: 500

Dataset rows used for training/evaluation: 300

Test accuracy: 70.7%

Balanced accuracy: 14.4%

Interpretation note: the underlying classroom dataset is small and imbalanced. Accuracy alone may overstate performance, so this report also shows balanced accuracy. The predictions should not be used as a clinical diagnosis.

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